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Differential entropy estimation with a Paretian kernel: Tail heaviness and smoothing

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ABSTRACT

Differential entropy extends the concept of entropy to continuous probability distributions, measuring the uncertainty associated with a continuous random variable. In financial data analysis, accurately estimating differential entropy is pivotal for understanding market dynamics and assessing risk. Traditional methods often fall short when dealing with the heavy-tailed distributions characteristic of financial returns. This paper introduces a novel approach to differential entropy estimation employing a Paretian kernel function adept at handling tail heaviness's intricacies. By incorporating an additional smoothing parameter, the Pareto exponent, our method offers flexibility in adjusting to light and heavy-tailed distributions. We compare our approach against established estimators through a comprehensive Monte Carlo simulation, demonstrating its superior performance in various scenarios. Applying our method to foreign exchange market data further illustrates its practical utility in identifying stochastic regimes and enhancing financial analysis. Our findings advocate for integrating the Paretian kernel estimator into the toolkit of financial analysts and researchers for a more nuanced understanding of market behavior.

1. Introduction

Differential entropy, a measure of uncertainty in continuous probability distributions in information theory, provides useful tools for analyzing financial data [1–11]. It is often estimated using statistical nonparametric methods when the underlying population distribution is unknown or complex [12–15]. Among the most widespread ones, the kernel density plug-in estimator (KDE) of the differential entropy is an approach where the probability density function for calculating entropy is estimated by smoothing sample data points through a convenient kernel function. As an alternative method, the k-nearest neighbor (K-NN) technique allows the estimation of differential entropy by analyzing the distances between each data point and its nearest neighbors, inferring density from these proximities. Finally, partitioning-type methods involve dividing the data space and estimating probabilities in each segment to compute differential entropy. Generally, they are competing estimators under some smoothness, tail, and peak conditions [12].

However, some issues arise when the underlying distributions are heavy-tailed. Regarding the KDE plug-in estimator, choosing a suitable kernel function depends on the adverse influence of tail behavior of the underlying population distribution [16,17]. The K-NN method can be sensitive to the tail behavior of the distribution because the distances between a point and its k nearest neighbors can be quite large, which may lead to higher bias. In partitioning methods, the spacing between samples within these

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partitions can affect the accuracy of the entropy estimation. However, consistent estimates can be produced from non-consistent spacing density estimates for large samples [12,18].

As heavy-tailed distributions are ubiquitous in financial data [19–24], the improvement of the differential entropy estimation technique is of fundamental importance. It is known that a heavy tail kernel may counteract the adverse influence of tail behavior [16,17]. Therefore, this paper suggests a flexible KDE plug-in estimator of the differential entropy using a Paretian kernel function, where Pareto's exponent accommodates tail heaviness. Thus, our approach allows us to deal with light and heavy tail situations.

In reviewing estimation methods, Ahmad and Lin [25] presented a nonparametric approach for estimating differential entropy via kernel density estimation, focusing on the robustness and consistency of entropy estimates across various distributions like Gamma, Weibull, and normal. Kraml et al. [26] developed X-Entropy, a tool for calculating entropy of distributions, particularly useful for dihedral angles in proteins, highlighting the efficiency of Gaussian kernel density estimation implemented in C++ and accessible via a Python frontend. Garcin [27] extended kernel density estimation to financial markets, introducing a novel bandwidth selection method that enhances market efficiency analysis, specifically for Bitcoin price returns. Lake [28] utilized kernel density estimation for the Rényi entropy family in biological and medical sciences, demonstrating its application in detecting abnormal cardiac rhythms via quadratic entropy estimation. These studies underline the diverse applications of kernel density estimation in scientific fields and its impact on statistical analysis and applied research.

Our novel approach builds on this foundation by employing a Paretian kernel function designed to handle the complexities of heavy-tailed distributions prevalent in financial data, a scenario less addressed by traditional kernel density estimation methods. By integrating an adjustable Pareto exponent, our method accommodates a wider range of distribution tail behaviors and enhances the precision of entropy estimates in these challenging contexts. This flexibility is particularly valuable for financial analysts and researchers, offering a subtler understanding of market dynamics and risk assessments.

The core motivation for our study is driven by the critical challenge of accurately estimating differential entropy in financial markets, where data distributions often exhibit heavy tails—attributes not adequately addressed by traditional entropy estimation methods. Financial datasets characteristically differ from those in more controlled experimental environments due to their distributions' complexity, unpredictability, and heavy-tailed nature. These features complicate traditional statistical analysis, making standard tools like kernel density estimation potentially misleading without adjustments for heavy tails.

Our research introduces a novel Paretian kernel approach designed to improve entropy estimation under these conditions. This approach addresses a practical gap in financial data analysis. By better estimating the differential entropy of such distributions, our method provides a more reliable foundation for analyzing market dynamics and assessing financial risk. These are critical components in the decision-making processes of financial analysts and economic policymakers, who require accurate tools to navigate the complexities of modern financial systems.

In particular, the motivation for our study arises from a critical examination of existing methods for estimating differential entropy, particularly within heavy-tailed financial data contexts. Prior research, such as the work by Ahmad and Lin [25], has demonstrated the applicability of kernel density estimation to light-tailed distributions. Yet, while methods like those introduced by Kraml et al. [26] and Garcin [27] have expanded kernel density estimation's usability in scientific and financial analyses, they do not fully address the unique challenges that financial markets' heavy-tailed distributions pose. Similarly, Lake [28] has applied kernel density estimation to medical sciences, underlining its broad utility and highlighting a gap in addressing financial data's specific characteristics.

Our research introduces a novel Paretian kernel approach, innovatively tailored to better capture the complexities of financial distributions. Unlike traditional kernel density estimation methods, our approach integrates an adjustable Pareto exponent, providing the flexibility to accurately model both light and heavy-tailed behaviors inherent in financial datasets. This methodological improvement is crucial for financial analysts and researchers who require precise risk assessment and market dynamics analysis tools.

Through Monte Carlo simulations and practical applications to foreign exchange market data, we show that our Paretian kernel estimator not only meets but exceeds the performance of existing estimators. We perform this Monte Carlo study to validate our results by considering the Student's t distribution, which exhibits heavy or light tails depending on its parameter. We compare mean squared errors, biases, and variances of KDE, K-NN, and sample spacing estimators.

We illustrate our method by analyzing daily US dollar prices in Swiss franc (CHF) and Brazilian real (BRL) currencies collected by the Federal Reserve Bank of New York, identifying different stochastic regimes using differential entropy estimates with Pareto's kernel function.

The remainder of the paper is organized as follows. Our Paretian kernel entropy estimator is introduced in Section 2. A Monte Carlo study is carried out in Section 3, which aims to compare our method with KDE, K-NN, and sample spacing estimators as benchmarks. We illustrate our approach in Section 4 using daily data from foreign exchange markets. Finally, Section 5 concludes the study.

2. Paretian kernel entropy estimator

Letting X be a continuous real-valued random variable governed by a density function f, its differential entropy is defined as

$$H(f) = -\int_{-\infty}^{\infty} f(x)\ln f(x)dx = -\mathbb{E}[\ln f(X)], \tag{1}$$

Table 1
Examples of light-tailed and heavy-tailed kernels, where $u \in \mathbb{R}$.

Distribution	Kernel	Tail
Gaussian	$K(u) = \frac{1}{\sqrt{2\pi}} \exp(-u^2/2)$	Light
Laplace	$K(u) = \frac{1}{2} \exp(-\lvert u \rvert)$	Light
Pareto	$K_\alpha(u) = \frac{\alpha}{2(1+\lvert u \rvert)^{\alpha+1}}$	Heavy ($\alpha \le 2$) Light ($2 < \alpha < \infty$) Dirac delta ($\alpha = \infty$)
Cauchy	$K(u) = \frac{1}{\pi(1+u^2)}$	Heavy

where $f(x)\ln f(x) \equiv 0$ when $f(x) = 0$. However, in a practical situation where we estimate f by $\hat{f}$, effectively we are dealing with the differential cross-entropy

$$H(\hat{f}) = -\int_{-\infty}^{\infty} f(x)\ln \hat{f}(x)dx = -\mathbb{E}[\ln \hat{f}(x)], \tag{2}$$

in which the expected uncertainty of $\hat{f}$ is obtained over the underlying distribution. The difference between $H(f)$ and $H(\hat{f})$ coincides with the Kullback–Leibler divergence from $\hat{f}$ to f,

$$D_{KL}(\hat{f}, f) = H(\hat{f}) - H(f) = \int_{\mathbb{R}} f(x)\ln \frac{f(x)}{\hat{f}(x)}dx. \tag{3}$$

Because $D_{KL}(\hat{f}, f)$ is a non-negative measure, the plug-in estimator $H(\hat{f})$ always tend to overestimate $H(f)$.

Now, given a random sample $X_1, \dots, X_n$ drawn from a population f, the kernel estimator of f is

$$\hat{f}_h(x) = \frac{1}{nh}\sum_{i=1}^{n} K\left(\frac{x - X_i}{h}\right), \tag{4}$$

where $h > 0$ is the smoothing bandwidth and K is a Kernel function. The evaluation of the integral in Eq. (3) is hard if $\hat{f}(x)$ is a kernel density estimator [12]. As Eq. (4) is a consistent estimator of the population differential entropy $H(f)$ [16,17,25], we consider the Kullback–Leibler loss as the empirical counterpart of (3) discussed by Hall and Morton [16,17], that is

$$\hat{D}_{KL}(\hat{f}_h, f) = \frac{1}{n}\sum_{j=1}^{n} \ln \frac{f(X_j)}{\hat{f}_h(X_j)}, \tag{5}$$

where the optimal bandwidth $\hat{h}$ can be found by minimizing (5) or, equivalently, minimize the empirical cross-entropy

$$\hat{H}(\hat{f}_h) = -\frac{1}{n}\sum_{j=1}^{n} \ln \hat{f}_h(X_j). \tag{6}$$

However, as $K(0)$ does not depend on h when $x = X_i$ in (4), leading to inconsistent results, we get the cross-validation loss function suggested by Hall [16], Habbema et al. [29], and Duin [30]

$$CV(h) = -\frac{1}{n}\sum_{j=1}^{n} \ln\left[\frac{1}{(n-1)h}\sum_{i\neq j} K\left(\frac{X_j - X_i}{h}\right)\right], \tag{7}$$

such that $\hat{h} = \arg\min_{h>0} CV(h)$ [16,30]. By following this approach, the cross-validation estimate given by

$$\hat{H}(\hat{f}_{\hat{h}}) = CV(\hat{h}) \tag{8}$$

is a consistent estimator of $H(f)$ depending on the tail properties of the kernel K and the underlying density f [12,17,31].

One of the contributions of this paper is to show empirically how the cross-validation loss function behaves depending on the tail properties of the kernel K with a heavy-tailed population f. For this purpose, we consider the Gaussian and Laplace as light-tailed kernels, the Cauchy as a heavy-tailed kernel, and the Pareto as a switchable light/heavy-tailed kernel (Table 1).

The Pareto's kernel is an interesting special case due to the exponent α. Here, we take the Pareto's exponent as an extra smoothing parameter, defining the cross-validation loss as $CV(h, \alpha)$ to get

$$(\hat{h}, \hat{\alpha}) = \underset{h>0,\alpha>0}{\arg\min}\, CV(h, \alpha).$$

This is a flexible and versatile kernel that can be either heavy-tailed ($\alpha \le 2$) or light-tailed ($\alpha > 2$). As $\alpha \to \infty$, the Pareto's kernel approaches the Dirac delta function. Moreover, it accommodates two different degrees of heaviness: (i) $\int_{\mathbb{R}} uK(u)du = \infty$ for $\alpha \le 1$; and (ii) $\int_{\mathbb{R}} uK(u)du < \infty$ and $\int_{\mathbb{R}} u^2K(u)du = \infty$, for $1 < \alpha \le 2$. Thus, the tail heaviness of Pareto's kernel is left to be data-driven.

Fig. 1 depicts how the $CV(h)$ function is influenced by the heaviness of the kernel function when estimating $H(f)$ of the Cauchy distribution. Discontinuities are observed when using Gaussian and Laplace's kernels, which are light-tailed. The opposite is observed with Cauchy and Pareto's ($\alpha = 1$) kernels. To confirm that heavy-tailed kernels help to deal with a heavy-tailed density f, we perform a Monte Carlo study in the next section.

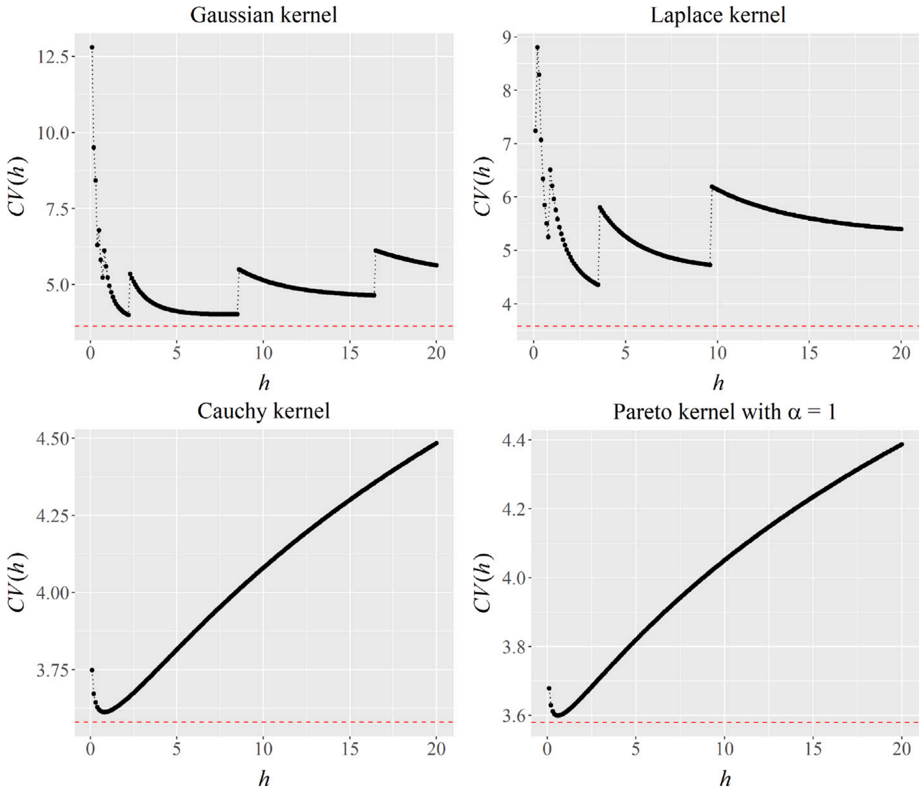

Fig. 1. Cross-validation loss function $CV(h)$ *versus* h to estimate the differential entropy of a Cauchy distribution with scale parameter $\sigma = 3$ using light-tailed (Gaussian and Pareto) and heavy-tailed (Cauchy and Laplace) kernel functions from samples of size $n = 500$. The dashed line indicates the true population entropy, $H(f) = \ln(4\pi\sigma)$. Discontinuities occur when using a light-tailed kernel to estimate the entropy of this Cauchy distribution.

3. A Monte Carlo study

We performed 500 Monte Carlo replications to assess our suggested approach. In each one, we generated a sample (n) of 30, 60, 200 and 500 deviations from a Student's t distribution with degrees of freedom (ν) ranging from 1 to 100, and we computed differential entropy estimates using the kernel functions from Table 1. We employ this distribution because it has heavier tails as ν decreases (Cauchy if $\nu = 1$) and the converse when ν increases (Gaussian if $\nu \to \infty$). When $\nu = 2$, we find a distribution with an undefined variance but a finite mean.

As a benchmark in this univariate study, we considered sample spacing (SS) and K-nearest Neighbor (KNN) estimates of the Student's t entropy $H(f)$, given by

$$H(f) = \frac{\nu+1}{2}\left[\psi\left(\frac{\nu+1}{2}\right) - \psi\left(\frac{\nu}{2}\right)\right] + \ln\left[\sqrt{\nu}B\left(\frac{\nu}{2}, \frac{1}{2}\right)\right],$$

where ψ means the digamma function and B is the beta function.

Tables 2–5 show the mean squared error, the variance, and the squared bias of the estimators from the Monte Carlo replications for different sample sizes. The estimators' performances are evaluated using three key metrics: mean squared error (MSE), variance ($\mathbb{V}[\hat{H}(\hat{f})]$), and bias ($B$). These metrics are crucial for understanding statistical estimators' strengths and limitations. The MSE quantifies the average squared difference between the estimated values and the actual value of the differential entropy. The variance of an estimator measures how much the estimates vary from one sample to another, and the bias measures how far the estimator's expected value is from the parameter's actual value. An estimator is biased if its expected value is not equal to the actual parameter value. The bias–variance decomposition encapsulates the relationship among MSE, variance, and bias, which states that MSE $= B^2 + \mathbb{V}[\hat{H}(\hat{f})]$. This fundamental equation illustrates the trade-off between bias and variance in statistical estimation. Thus, balancing bias and variance is crucial for developing good estimators.

Pareto's kernel generally offers the lowest mean squared errors among the competing estimators for all ν, closely followed by the sample spacing and Cauchy's kernel entropy estimators. However, NND estimator performed poorly (Fig. 2, left), although it provides very low bias (Tables 2–5). Fig. 2 (right) illustrates how the average $\bar{\alpha}$ evolves as a function of ν. As expected, $\hat{\alpha}$ increases as the distribution heaviness decreases (ν increases).

Table 2
Estimating differential entropy of an univariate Student's t distribution with v degrees of freedom with sample size (n) of 30: Mean squared error (MSE), variance ($\mathbb{V}[\hat{H}(\hat{f})]$), bias ($B$) from 500 Monte Carlo replications of different kernel entropy estimators, and NND and SS estimators as benchmarks.

v	Measures	Gaussian	Laplace	Cauchy	Pareto	NND	SS
1	MSE	1.871745	1.039242	**0.147495**	**0.142217**	0.167602	**0.140230**
	$\mathbb{V}[\hat{H}(\hat{f})]$	1.051349	0.703585	**0.128384**	**0.126923**	0.167396	**0.119128**
	B	0.905757	0.579359	0.138240	0.123672	**−0.014358**	−0.145265
2	MSE	0.301139	0.132805	**0.070143**	**0.071035**	0.133951	0.095869
	$\mathbb{V}[\hat{H}(\hat{f})]$	0.238220	0.114313	**0.062777**	**0.067484**	0.133691	0.079704
	B	0.250835	0.135987	0.085823	0.059589	**−0.016120**	−0.127145
3	MSE	0.092499	0.060081	**0.051705**	**0.049458**	0.107892	0.075273
	$\mathbb{V}[\hat{H}(\hat{f})]$	0.083407	0.056286	**0.045496**	**0.048100**	0.107471	0.060610
	B	0.095354	0.061597	0.078793	0.036849	**−0.020519**	−0.121088
4	MSE	0.063539	0.046281	0.046847	**0.041012**	0.106757	0.066903
	$\mathbb{V}[\hat{H}(\hat{f})]$	0.056403	0.042111	**0.038029**	**0.038641**	0.106737	0.055840
	B	0.084479	0.064577	0.093905	**0.048690**	**−0.004563**	−0.105180
5	MSE	0.043190	0.035130	0.039072	**0.033143**	0.092139	0.058305
	$\mathbb{V}[\hat{H}(\hat{f})]$	0.039520	**0.032632**	**0.031296**	**0.031570**	0.092106	0.047636
	B	0.060582	0.049977	0.088181	0.039666	**−0.005767**	−0.103290
10	MSE	0.030654	**0.029702**	0.038713	**0.029402**	0.081777	0.048630
	$\mathbb{V}[\hat{H}(\hat{f})]$	0.027928	**0.026767**	**0.028059**	**0.026992**	0.081775	0.040888
	B	0.052209	0.054181	0.103220	0.049090	**−0.001319**	−0.087987
15	MSE	**0.023968**	**0.023895**	0.032793	**0.023924**	0.084789	0.046880
	$\mathbb{V}[\hat{H}(\hat{f})]$	**0.022572**	**0.021984**	**0.023361**	**0.022317**	0.084300	0.040370
	B	0.037362	0.043722	0.097114	0.040095	**0.022109**	−0.080686
20	MSE	**0.021406**	**0.021687**	0.030099	**0.021704**	0.093718	0.051745
	$\mathbb{V}[\hat{H}(\hat{f})]$	**0.020677**	**0.020506**	**0.022180**	**0.020733**	0.093166	0.040672
	B	**0.026999**	0.034360	0.088988	0.031172	**−0.023485**	−0.105231
50	MSE	**0.019329**	**0.021065**	0.032546	**0.021354**	0.090133	0.048845
	$\mathbb{V}[\hat{H}(\hat{f})]$	**0.017519**	**0.018537**	0.021618	**0.019165**	0.090067	0.040523
	B	0.042545	0.050277	0.104536	0.046789	**−0.008119**	−0.091226
100	MSE	**0.019114**	**0.020215**	0.030741	**0.020602**	0.081790	0.045024
	$\mathbb{V}[\hat{H}(\hat{f})]$	**0.018049**	**0.018454**	**0.020913**	**0.019063**	0.081788	0.037437
	B	0.032636	0.041966	0.099139	0.039234	**0.001469**	−0.087105

Table 3
The same as in Table 2, with $n = 60$.

v	Measures	Gaussian	Laplace	Cauchy	Pareto	NND	SS
1	MSE	1.972929	0.947731	**0.073669**	**0.070149**	0.085341	**0.070323**
	$\mathbb{V}[\hat{H}(\hat{f})]$	0.825971	0.535231	**0.062680**	**0.060351**	0.085215	**0.060827**
	B	1.070961	0.642262	0.104829	0.098982	**−0.011222**	−0.097447
2	MSE	0.263333	0.081895	**0.034361**	**0.034856**	0.064001	0.045408
	$\mathbb{V}[\hat{H}(\hat{f})]$	0.184488	0.062880	**0.029761**	**0.031956**	0.063565	0.038512
	B	0.280793	0.137896	0.067820	0.053851	**−0.020880**	−0.083042
3	MSE	0.061122	0.030743	**0.023150**	**0.022043**	0.054686	0.035451
	$\mathbb{V}[\hat{H}(\hat{f})]$	0.051147	0.027608	**0.020080**	**0.021148**	0.054208	0.028531
	B	0.099878	0.055992	0.055413	0.029928	**−0.021862**	−0.083188
4	MSE	0.034608	0.021970	0.021701	**0.018882**	0.047251	0.029647
	$\mathbb{V}[\hat{H}(\hat{f})]$	0.029358	0.019472	**0.017108**	**0.017579**	0.047230	0.025458
	B	0.072453	0.049978	0.067776	0.036099	**−0.004670**	−0.064719
5	MSE	0.023823	0.017950	0.019301	**0.016201**	0.046830	0.027850
	$\mathbb{V}[\hat{H}(\hat{f})]$	0.021011	0.016346	**0.014945**	**0.015217**	0.046710	0.023430
	B	0.053025	0.040056	0.065994	0.031363	**−0.010961**	−0.066485
10	MSE	0.015038	**0.014311**	0.018756	**0.014091**	0.045330	0.025312
	$\mathbb{V}[\hat{H}(\hat{f})]$	**0.013934**	**0.013216**	**0.013562**	**0.013163**	0.045315	0.021976
	B	0.033229	0.033089	0.072066	0.030464	**−0.003887**	−0.057752
15	MSE	**0.012690**	**0.012539**	0.017112	**0.012417**	0.045469	0.024823
	$\mathbb{V}[\hat{H}(\hat{f})]$	**0.012041**	**0.011746**	**0.012295**	**0.011731**	0.045464	0.021902
	B	0.025470	0.028158	0.069406	0.026184	**0.002347**	−0.054040

(continued on next page)

Table 3 (*continued*).

v	Measures	Gaussian	Laplace	Cauchy	Pareto	NND	SS
20	MSE	**0.011401**	**0.011812**	0.017511	**0.011902**	0.047874	0.024483
	$\mathbb{V}[\hat{H}(\hat{f})]$	**0.010646**	**0.010852**	**0.012003**	**0.011045**	.047821	0.021463
	B	0.027471	0.030987	0.074214	0.029276	−0.007277	−0.054946
50	MSE	**0.010095**	**0.010423**	0.015653	**0.010533**	0.046752	0.024586
	$\mathbb{V}[\hat{H}(\hat{f})]$	**0.009578**	**0.009649**	**0.010369**	**0.009839**	0.046558	0.020946
	B	0.022731	0.027823	0.072695	0.026351	**−0.013930**	−0.060336
100	MSE	**0.009310**	**0.009735**	0.015380	**0.009750**	0.043105	0.021878
	$\mathbb{V}[\hat{H}(\hat{f})]$	**0.008739**	**0.008869**	**0.009791**	**0.008948**	0.043098	0.019040
	B	0.023896	0.029426	0.074763	0.028313	**−0.002513**	−0.053276

Table 4
The same as in Tables 2 and 3, with $n = 200$.

v	Measures	Gaussian	Laplace	Cauchy	Pareto	NND	SS
20	MSE	2.056461	0.972367	**0.024827**	**0.023732**	0.028541	**0.022370**
	$\mathbb{V}[\hat{H}(\hat{f})]$	0.555130	0.441131	**0.020103**	**0.019312**	0.028473	0.020319
	B	1.225288	0.728859	0.068729	0.066486	**−0.008253**	−0.045285
2	MSE	0.220813	0.036424	**0.009879**	**0.009887**	0.019615	0.012871
	$\mathbb{V}[\hat{H}(\hat{f})]$	0.131443	0.022170	**0.008387**	**0.008679**	0.019593	0.011724
	B	0.298947	0.119393	0.038624	0.034755	**−0.004646**	−0.033872
3	MSE	0.047055	0.012427	**0.007544**	**0.007240**	0.017364	0.010575
	$\mathbb{V}[\hat{H}(\hat{f})]$	0.035882	0.010177	**0.006589**	**0.006874**	0.017364	0.009604
	B	0.105701	0.047427	0.030896	0.019130	**0.000675**	−0.031147
4	MSE	0.013472	0.007205	0.006768	**0.005833**	0.015219	0.008752
	$\mathbb{V}[\hat{H}(\hat{f})]$	0.010363	0.006018	**0.005213**	**0.005302**	0.015185	0.008324
	B	0.055753	0.034445	0.039442	0.023031	**0.005825**	−0.020675
5	MSE	0.007830	0.005548	0.006063	**0.004986**	0.015011	0.008127
	$\mathbb{V}[\hat{H}(\hat{f})]$	0.006338	**0.004840**	**0.004572**	**0.004607**	0.014966	0.007383
	B	0.038618	0.026623	0.038613	0.019482	**−0.006642**	−0.027271
10	MSE	**0.004617**	**0.004283**	0.005684	**0.004145**	0.014192	0.007519
	$\mathbb{V}[\hat{H}(\hat{f})]$	0.004170	**0.003898**	**0.003862**	**0.003824**	0.014191	0.007089
	B	0.021137	0.019624	0.042694	0.017913	**0.000820**	−0.020737
15	MSE	**0.003791**	**0.003681**	0.005332	**0.003635**	0.014281	0.006858
	$\mathbb{V}[\hat{H}(\hat{f})]$	**0.003447**	**0.003349**	**0.003458**	**0.003335**	0.014280	0.006429
	B	0.018535	0.018212	0.043287	0.017309	**−0.000600**	−0.020723
20	MSE	**0.003258**	**0.003225**	0.004928	**0.003193**	0.014175	0.006827
	$\mathbb{V}[\hat{H}(\hat{f})]$	**0.003010**	**0.002959**	**0.003074**	**0.002951**	0.014147	0.006311
	B	0.015766	0.016338	0.043058	0.015570	**−0.005284**	−0.022703
50	MSE	**0.002831**	**0.002895**	0.004837	**0.002896**	0.012913	0.006024
	$\mathbb{V}[\hat{H}(\hat{f})]$	**0.002557**	**0.002578**	**0.002769**	**0.002593**	0.012890	0.005728
	B	0.016546	0.017819	0.045466	0.017400	**0.004854**	−0.017223
100	MSE	**0.002833**	**0.002858**	0.004513	**0.002859**	0.014286	0.006583
	$\mathbb{V}[\hat{H}(\hat{f})]$	**0.002648**	**0.002640**	**0.002740**	**0.002654**	0.014266	0.006233
	B	0.013601	0.014747	0.042112	0.014302	**0.004500**	−0.018706

4. Illustration

Different statistical models can show heavy tails in foreign exchange rate distributions [32] For example, using stable Paretian models [33] and generalized t distributions [34], as well as taking into account long memory [35] and Hurst exponents [36]. In our illustration, we aim to show how our approach may be useful in detecting different stochastic regimes in the historical evolution of an exchange rate. Consider the following as an example of a change in the exchange rate regime. In March 1973, several nations abandoned efforts to maintain the system of fixed exchange rates, beginning a period of transition from fixed to floating (market-determined) exchange rates between the US dollar and other major currencies.

We choose additional examples involving the Swiss Franc (CHF) and the Brazilian Real (BRL). Although the CHF is often referred to as a hard currency, on 15 January 2015, the Swiss National Bank suddenly announced that it would no longer hold the CHF at a fixed exchange rate against the euro. This resulted in a sharp drop in the rate from that day onwards [37]. In contrast, the BRL commonly experiences different regimes. The first began when Brazil implemented the Real Plan in 1993 as a monetary exchange rate-based stabilization policy, where the central element was an exchange rate anchor to reduce chronic inflation. The BRL currency was introduced in July 1994 under an exchange rate band strategy until 13 January 1999, when the first real currency crisis occurred. Just after that, the real-dollar rate was let to be market-determined. The BRL experienced a log-periodic bubble (28 May 2002–14 Jan 2003), possibly because of uncertainties in the Brazilian presidential election and the economic policy to be adopted by Lula,

Table 5
The same as in Tables 2–4, with $n = 500$.

v	Measures	Gaussian	Laplace	Cauchy	Pareto	NND	SS
1	MSE	0.412719	0.641096	0.011115	**0.010730**	0.011699	**0.009152**
	$\mathbb{V}[\hat{H}(\hat{f})]$	0.087178	0.218827	**0.008426**	**0.008142**	0.011698	0.008751
	B	0.570562	0.649823	0.051854	0.050866	**−0.001060**	−0.020018
2	MSE	0.109086	0.020924	**0.003946**	**0.003965**	0.007454	0.004392
	$\mathbb{V}[\hat{H}(\hat{f})]$	0.052868	0.011148	**0.003133**	**0.003244**	0.007418	0.004269
	B	0.237104	0.098872	0.028508	0.026853	**0.005975**	−0.011106
3	MSE	0.022403	0.005356	0.003139	**0.002872**	0.006675	0.003738
	$\mathbb{V}[\hat{H}(\hat{f})]$	0.013795	0.003496	**0.002342**	**0.002416**	0.006642	0.003671
	B	0.092778	0.043128	0.028231	0.021362	**0.005727**	−0.008187
4	MSE	0.006678	0.002625	0.002320	**0.002006**	0.006418	0.003446
	$\mathbb{V}[\hat{H}(\hat{f})]$	0.005023	0.002214	**0.001868**	**0.001890**	0.006396	0.003155
	B	0.040685	0.020283	0.021253	0.010763	**−0.004760**	−0.017076
5	MSE	0.003356	0.002017	0.002174	**0.001735**	0.006271	0.003170
	$\mathbb{V}[\hat{H}(\hat{f})]$	0.002538	**0.001702**	**0.001561**	**0.001586**	0.006261	0.003021
	B	0.028593	0.017763	0.024749	0.012213	**−0.003126**	−0.012222
10	MSE	0.001600	0.001521	0.002097	**0.001493**	0.005846	0.002784
	$\mathbb{V}[\hat{H}(\hat{f})]$	**0.001462**	**0.001411**	**0.001411**	**0.001404**	0.005846	0.002678
	B	0.011765	0.010462	0.026201	0.009435	**0.000573**	−0.010292
15	MSE	0.001310	0.001307	0.001986	**0.001304**	0.005027	0.002321
	$\mathbb{V}[\hat{H}(\hat{f})]$	**0.001212**	**0.001210**	**0.001247**	**0.001218**	0.005024	0.002251
	B	0.009870	0.009843	0.027183	0.009293	**0.001551**	−0.008403
20	MSE	0.001305	**0.001279**	0.001939	**0.001271**	0.005441	0.002608
	$\mathbb{V}[\hat{H}(\hat{f})]$	**0.001227**	**0.001198**	**0.001210**	**0.001197**	0.005421	0.002451
	B	0.008821	0.009002	0.027003	0.008611	**−0.004441**	−0.012568
50	MSE	**0.001217**	**0.001225**	0.001942	**0.001224**	0.005545	0.002572
	$\mathbb{V}[\hat{H}(\hat{f})]$	**0.001164**	**0.001160**	**0.001195**	**0.001162**	0.005531	0.002433
	B	0.007245	0.008062	0.027336	0.007907	**−0.003793**	−0.011752
100	MSE	**0.001098**	**0.001105**	0.001820	**0.001102**	0.005222	0.002311
	$\mathbb{V}[\hat{H}(\hat{f})]$	**0.001049**	**0.001042**	**0.001061**	**0.001041**	0.005220	0.002201
	B	0.007041	0.007941	0.027542	0.007846	**−0.001626**	−0.010494

the elected president [38]. Other factors occurred, such as the impeachment of President Dilma Roussef, the period of austerity during Michel Temer's short government, and the transition to Jair Bolsonaro's government, marking the country as a rollercoaster of economic and monetary policies.

Now, ignoring such historical facts, we apply our approach to detect them using only our data-driven approach with the following differential entropy estimation through the Pareto kernel. Given a time series of exchange rates $\{P_t\}$, we obtain the log-return as usual, $X_t = \ln P_t/P_{t-1}$, where t is a time index varying from 2 to n, where n denotes the sample size. In our illustration, we divided the time series of returns exactly into blocks of size b, so that $n = b \times m$. We deal with m time series of size b. For each block, using Pareto's kernel, we calculate the differential entropy with the estimated α, and we classify the block according to its magnitude. From Table 1. A heavy-tailed kernel occurs if $\ln \alpha \leq \ln 2$. Empirically, we find it reasonable to classify $\ln 2 < \ln \alpha \leq \ln 5$ as the light-tailed kernel case and $\ln \alpha > 5$ as an approximation of the Dirac's delta kernel.

The interpretation is straightforward. Price movements are random and unpredictable in perfectly efficient markets because they instantly incorporate all available information, leaving no room for predictable patterns or arbitrage opportunities. Several alternative methodologies can be considered when examining the efficiency of foreign exchange rates, such as Shannon entropy [39, 40], Hurst exponent [36], detrending moving average [41], mean reversion [42], correlation structure [39], and algorithmic complexity [43]. In our approach, when α is high, the kernel function at a certain point X_t depends almost nothing on the other sample points of the data block. We interpret it as economic efficiency because of a high degree of randomness in the data series in measuring entropy. At the opposite extreme, a small parameter ($\alpha \leq 2$) means that the kernel function at a certain point X_t depends on almost all the other sample points of the block. We interpret it as a lack of economic efficiency, with a high degree of dependence on the entropy estimation. Thus, smaller α's are related to turbulent periods, while higher α means regular periods with smaller tails. Interestingly, higher/lower α does not imply more or less uncertainty or volatility because it is a kernel's parameter.

We interpret certain outcomes as indications of a lack of market efficiency based on differential entropy estimates that suggest higher dependency among return series in periods identified as less efficient. This interpretation stems from the theoretical underpinning that price movements should be entirely unpredictable and purely random in a perfectly efficient market, reflecting all available information instantaneously. Thus, higher differential entropy values, indicating lower predictability and greater randomness, would suggest greater market efficiency.

Our Paretian kernel method, designed to handle heavy-tailed distributions common in financial markets, can provide insights into market efficiency. The method's sensitivity to tail behavior makes it particularly suited to detecting phases where market data does not behave according to the efficient market hypothesis. By analyzing the tail-dependency of financial returns, our method can

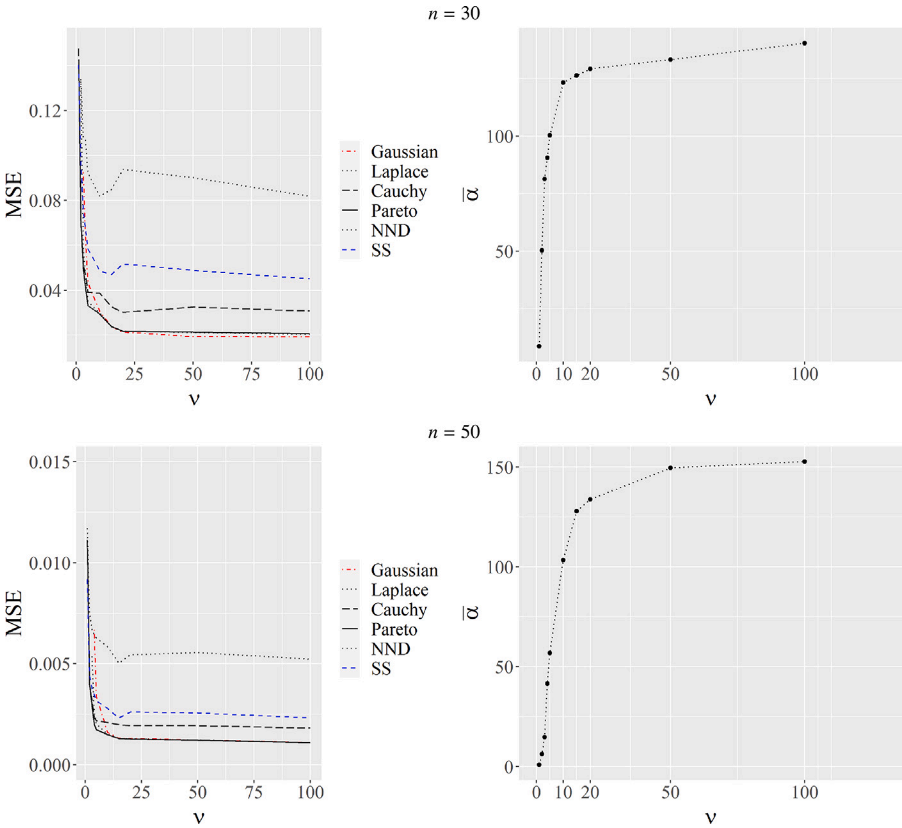

Fig. 2. Left: Mean squared errors (MSE) of kernel differential entropy estimates of a Student's t distribution with ν degrees of freedom from 500 Monte Carlo replications for $n = 30$ (top) and $n = 500$ (bottom). **Right:** Average values of $\hat{\alpha}$ for different values of ν when using Pareto's kernel from the univariate Monte Carlo experiments (Tables 2 and 5).

reveal periods of potential market inefficiency characterized by predictability in price movements or an abnormal accumulation of price information.

To further explore this capability, we propose a systematic study where the Paretian kernel-based entropy estimation is directly applied as a test for market efficiency. We can analyze the entropy levels over time and correlate these with known market events or regulatory changes expected to affect market efficiency. A significant change in entropy in response to such events could validate the utility of our approach as a market efficiency test. Moreover, employing our method in a comparative analysis with other traditional market efficiency tests could highlight its unique advantages or potential limitations, providing a comprehensive view of its applicability in financial analysis.

The data sets employed in this illustration were taken from the Federal Reserve website (http://www.federalreserve.gov/releases/H10/hist/). They refer to a currency value in US dollar terms collected by the Federal Reserve Bank of New York from a sample of market participants. We ignore gaps from weekends and holidays, concentrating our analysis on trading days.

We choose $b = 60$, which gives $m = 221$ and 121 for the CHF and BRL data, respectively. Fig. 3 shows the BRL/USD and CHF/USD time series of exchange rates (CHF from 5-Jan-1971 to 16-Nov-2023 and BRL from 3-Jan-1995 to 12-Dec-2023). Interestingly, although the kernel types are obtained from blocks of log returns (Fig. 4), we may find a general picture consistent with some known historical facts that affected price movements. Fig. 5 depicts differential entropy estimates against sample variances from each data block. It shows that some entropies estimated with low α (heavy tail kernel) are further away from the Gaussian line. However, we also find blocks with small variances and entropies with small α. It reinforces that tail heaviness, uncertainty, and volatility are not synonymous, which must be considered when measuring financial risks.

Table 6 lists the heavy-tailed blocks (13 CHF and 21 BRL blocks), which align with some facts described at the beginning of this section. Finally, Fig. 6 shows differential entropy estimates against estimated values of $\ln\alpha$, revealing the existence of three kernel types. The heavy-tailed case is established theoretically because the Pareto distribution has no variance when $\ln\alpha \le \ln 2$. Classifying $\ln\alpha > 5$ as an approximation of Dirac's delta kernel is reasonable because a cluster of points is formed after $\ln\alpha = 5$.

5. Conclusion

This study underscores the crucial role of tail behavior in estimating differential entropy, especially within financial data analysis. Our findings advocate adopting a Paretian kernel-based estimator as a robust tool that accommodates the complexities inherent in

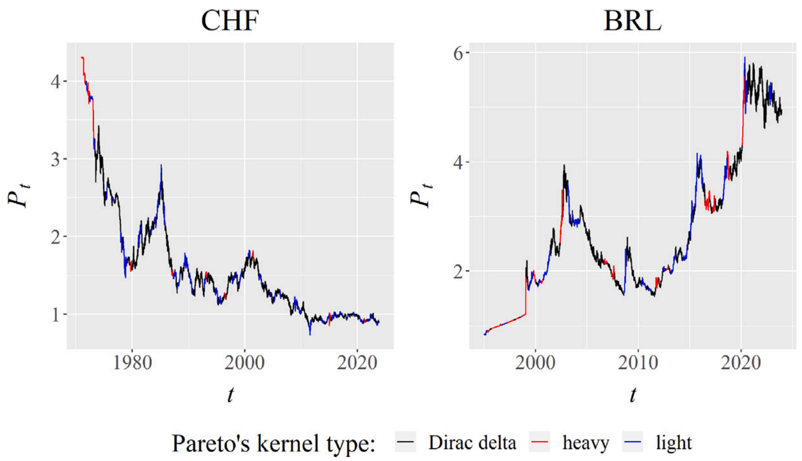

Fig. 3. Historical evolution of foreign exchange rates against the US dollar. Swiss franc (CHF): n = 13,260 observations from 5-Jan-1971 to 16-Nov-2023. Brazilian real (BRL): n = 7260 from 4-Jan-1995 to 13-Dec-2023. We find Pareto's kernel types suggesting different regimes generally coherent with known historical facts such as the 1970s exchange rate system change (CHF) and the Real Plan in the early 1990s (BRL).

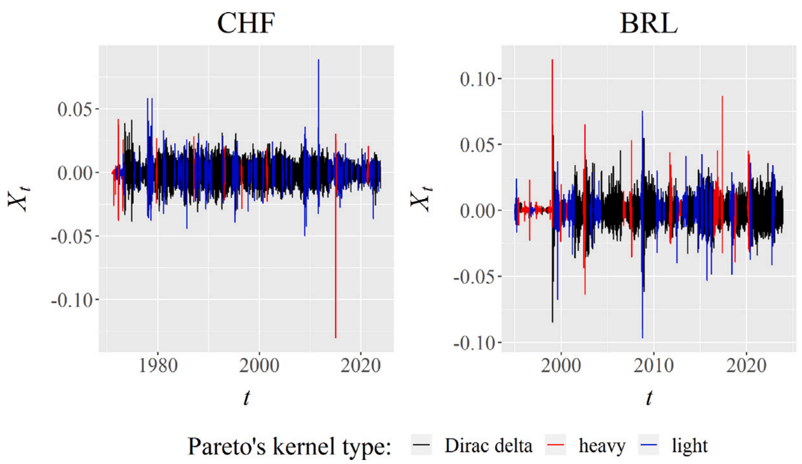

Fig. 4. Log returns from data shown in Fig. 3, with the Pareto's kernel types suggesting different stochastic regimes.

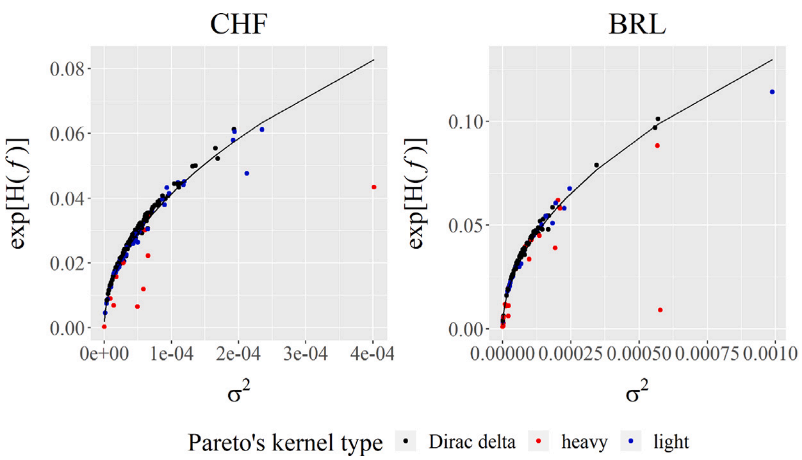

Fig. 5. m estimated differential entropies *versus* sample variances calculated for each block of size b = 60, where m = 221 for CHF and m = 121 for BRL data. The red line refers to the Gaussian differential entropy given by $H(f) = 0.5\ln(2\pi\sigma^2) + 0.5$.

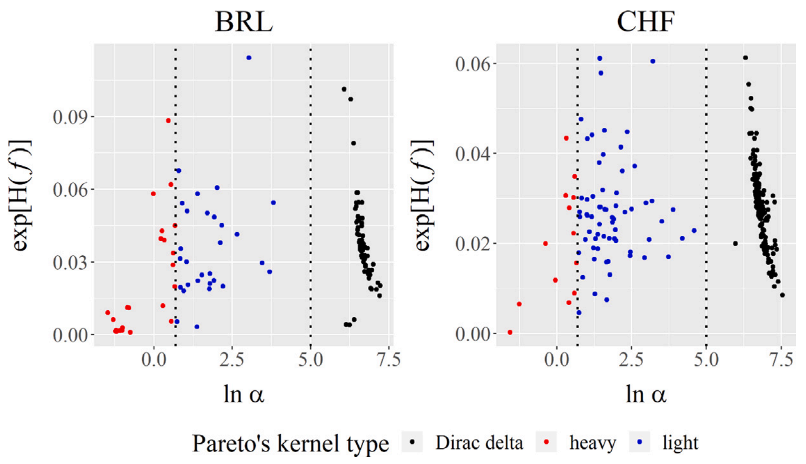

Fig. 6. m estimated differential entropies *versus* estimated $\ln\alpha$ for each block of size $b = 60$, where $m = 221$ for CHF and $m = 121$ for BRL data. We find three clusters of α discriminated by dashed vertical lines: $\ln\alpha \le \ln 2$ (heavy-tailed kernel), $\ln 2 < \ln\alpha \le 5$ (light-tailed kernel) and we consider $\ln\alpha > 5$ as an approximation of the Dirac's delta kernel.

Table 6
Blocks with differential entropies estimated with heavy-tailed kernel.

Block	Period	$\hat{\alpha}$	$\hat{h}$	$\hat{H}(\hat{f}_{\hat{h}})$
Swiss franc (CHF)				
1	1971-01-05–1971-03-31	0.21	5.5×10^{-9}	−8.35
2	1971-04-01–1971-06-24	1.50	4.0×10^{-4}	−4.99
3	1971-06-25–1971-09-27	1.80	8.9×10^{-4}	−4.72
4	1971-12-28–1972-03-21	0.95	2.5×10^{-4}	−4.44
5	1972-03-22–1972-06-14	0.29	1.3×10^{-6}	−5.03
6	1972-12-08–1973-03-07	1.75	2.1×10^{-3}	−3.81
7	1979-08-23–1979-11-20	1.34	1.2×10^{-3}	−3.49
8	1987-01-29–1987-04-23	1.75	1.9×10^{-3}	−3.50
9	1993-01-15–1993-04-12	1.81	1.7×10^{-6}	−3.36
10	1996-05-16–1996-08-09	0.68	9.5×10^{-5}	−3.91
11	2001-05-18–2001-08-13	1.51	1.3×10^{-3}	−3.58
12	2014-12-30–2015-03-26	1.37	2.8×10^{-3}	−3.14
13	2021-03-31–2021-06-23	1.92	1.5×10^{-3}	−4.16
Brazilian real (BRL)				
Block	Period	$\hat{\alpha}$	$\hat{h}$	$\hat{H}(\hat{f}_{\hat{h}})$
1	1995-03-28–1995-06-20	0.43	8.8×10^{-6}	−4.49
2	1995-09-15–1995-12-12	0.33	4.7×10^{-7}	−6.39
3	1995-12-13–1996-03-15	0.37	5.4×10^{-7}	−5.88
4	1996-03-18–1996-06-10	0.30	3.4×10^{-7}	−6.30
5	1996-06-11–1996-09-04	0.27	1.2×10^{-6}	−5.09
6	1997-03-03–1997-05-23	0.29	2.8×10^{-7}	−6.55
7	1997-05-27–1997-08-19	0.30	2.9×10^{-7}	−6.62
8	1997-08-20–1997-11-14	0.36	7.0×10^{-7}	−6.31
9	1998-02-13–1998-05-08	0.47	8.0×10^{-7}	−6.96
10	1998-08-04–1998-10-28	1.74	5.0×10^{-4}	−5.21
11	1998-10-29–1999-01-27	0.23	6.0×10^{-7}	−4.71
12	1999-10-14–2000-01-07	1.84	2.2×10^{-3}	−3.55
13	2000-06-29–2000-09-22	0.45	1.2×10^{-5}	−4.50
14	2002-05-30–2002-08-22	1.59	3.1×10^{-3}	−2.43
15	2006-09-14–2006-12-08	1.94	1.2×10^{-3}	−3.92
16	2007-06-04–2007-08-27	1.95	4.2×10^{-3}	−3.10
17	2011-09-15–2011-12-12	1.73	2.9×10^{-3}	−2.78
18	2012-08-29–2012-11-26	1.33	2.7×10^{-4}	−4.44
19	2016-06-29–2016-09-22	1.25	7.8×10^{-4}	−3.23
20	2016-09-23–2016-12-20	1.85	2.7×10^{-3}	−3.39
21	2017-03-22–2017-06-14	1.39	1.8×10^{-3}	−3.24
22	2018-08-24–2018-11-20	1.29	1.1×10^{-3}	−3.15
23	2020-02-10–2020-05-04	0.98	7.2×10^{-4}	−2.84

financial distributions. This method enhances the accuracy of differential entropy estimation and provides a nuanced understanding of financial market dynamics through the lens of entropy. Future research could explore integrating this approach with other financial modeling techniques to further unravel the intricate behaviors of markets and enhance risk assessment methodologies. Emphasizing the adaptability and efficacy of the Paretian kernel, our research contributes to the advancement of statistical tools in financial analysis. The R codes and functions for estimating differential entropy using Pareto's kernel and data sets are available on Figshare (https://doi.org/10.6084/m9.figshare.25688094.v1).

CRediT authorship contribution statement

Raul Matsushita: Writing – original draft, Validation, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **Helena Brandão:** Methodology, Formal analysis, Conceptualization. **Iuri Nobre:** Writing – original draft, Investigation, Conceptualization. **Sergio Da Silva:** Writing – review & editing, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The R codes and functions for estimating differential entropy using Pareto's kernel and data sets are available on Figshare (https://doi.org/10.6084/m9.figshare.25688094.v1)

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